JOHN KOCHENSPARGER

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SUMMARY

Electrical engineer who takes hardware failures from first symptom to verified fix. Electronics Engineer at Air Force Metrology and Calibration (AFMETCAL), reviewing drawings, specifications, and test plans for incoming systems. Two years of cryocooler R&D at AMETEK Sunpower tracing ESD and production failures to root cause and retesting until they passed. Led a rocket club avionics team, co-founded a hardware startup, and worked the floor of two asphalt emulsion plants.

EXPERIENCE

Air Force Metrology and Calibration (AFMETCAL) | Heath, OH — Sep 2026 – Present
Electronics Engineer, NH-0855-2

- Review engineering drawings, design specifications, and test plans for new calibration systems the Air Force is acquiring.
- Trace failures in test equipment and calibration standards to systemic root causes and engineer reliability modifications.
- Review, update, and validate calibration Technical Orders for microwave, RF, electro-optical, and voltage systems.
- Serve as a subject matter expert for Air Force calibration labs (PMELs) worldwide, resolving measurement anomalies.
- Automate calibration procedures; work on intrinsic standards and R&D acquisitions that close measurement gaps.

AMETEK Sunpower Inc. | Athens, OH — May 2024 – Apr 2026
Electrical Engineering R&D Intern | Free-piston Stirling cryocoolers

- Ran IEC/EN 61000-4-2 ESD compliance testing on two cryocooler controller systems with a 16.5 kV simulator: captured overcurrent events on an oscilloscope, traced each failure, modified the PCB, and retested until each controller passed.
- Isolated the startup conditions that made a production CryoTel DS30 fail and traced the source with oscilloscope captures.
- Built a Python test-automation framework from scratch (serial I/O, JSON-defined procedures, emulator profile) covering 358 procedures: auto-detects board revision, firmware, and COM settings, validates configuration and sensor calibration, and logs for QA, RMA, and MRB; cut per-procedure time by 95%.
- Designed a 600 W test load matching CryoTel GT impedance and operating frequency for controller validation; sized the heatsink and fan from steady-state thermal analysis.
- Wrote test reports, user manuals, procedures, and BOMs; tracked component lifecycles and replaced NFND/obsolete parts.

Eye Proxy LLC | Athens, OH — Oct 2025 – Present
Co-Founder

- Co-founded a two-person company building a covert dashcam disguised as a parking pass (patent pending); lead patent work, documentation, firmware, and business development while my partner leads hardware and procurement.
- Project manager for the senior capstone that started it: ran weekly meetings, minutes, and action items through SRR, PDR, and CDR design reviews, and presented the working prototype at two expos.
- Designed the enclosure across three revisions and ~10 printed prototypes; caught supplier CAD models that did not match the purchased parts and corrected the design to caliper measurements.
- Hand-reworked a fine-pitch module with hot-air reflow beside 0201 parts; in parallel, built a development-board proof of concept that validated the data-rate target and kept firmware unblocked.

Asphalt Materials Inc. | Marietta, OH — May 2023 – Aug 2023
Plant Operator Intern

- Wrote 200+ valve alignment procedures covering most plant material transfers, in line with OSHA standards.
- Assisted a controls engineer in integrating a VFD with a PLC to control asphalt pump flow; reworked the mill room layout.
- Rotated between two asphalt emulsion plants: weighed loads, loaded asphalt onto trucks, and measured water parameters.

EDUCATION

Ohio University | Athens, OH — May 2026
Bachelor of Science in Electrical Engineering (BSEE) | IEEE Member

- **Avionics Lead, Ohio University Rocket Club:** Led the club's avionics hardware team: selected flight hardware, wrote firmware logging flight data on five L1 launches, and built a Python pipeline for 200k to 300k sensor readings per flight.

SKILLS

Field & Analysis: Root cause analysis; drawing, spec, and test plan review; ESD testing (IEC 61000-4-2); technical reports
Leadership & Business: Team leadership; SRR/PDR/CDR design reviews; patent work; business development
Metrology: Calibration Technical Orders and standards; microwave, RF, electro-optical, and voltage measurement
Hands-On: Lead-free soldering (iron, hot air), PCB rework, cable fabrication, fiber splicing, oscilloscopes
Design: Altium, KiCad, Multisim, AutoCAD, SolidWorks, Autodesk Inventor
Programming: Python (pyserial, test automation), C, C++, MATLAB, Git, LaTeX; Windows, Linux, macOS